A signature for pan-cancer prognosis based on neutrophil extracellular traps

Yi Zhang, Liping Guo, Qichen Dai, Bingqing Shang, Ting Xiao, Xuebing Di, Kaitai Zhang, Lin Feng, Jianzhong Shou & Yipeng Wang

National Cancer Center / Cancer Hospital, CAMS & PUMC · Peking University People's Hospital · collaborating centers

J Immunother Cancer 2022 · DOI 10.1136/jitc-2021-004210

Institution logo

19 genes in the NETs signature

8,739 TCGA pan-cancer cases

4 IHC validation cohorts

Why NETs?

Neutrophil extracellular traps (NETs) can support tumor progression and dissemination. The study asked whether transcriptomic NETs-formation potential could stratify prognosis across cancer types.

Evidence: Abstract and Introduction, pp. 1-2

Model construction

1. 69 initial NETs biomarkers
2. LASSO retained 24 genes
3. Correlation pruning removed 5 genes
4. 19-gene Cox-derived NETs score

DATA FOUNDATION

TCGA: n=8,739, 32 tumor types
Training: n=6,117 · Test: n=2,622
CGGA: n=651 · METABRIC: n=1,868 · GEO: n=2,459
IHC: LUAD n=58 · COAD n=93 · KIRC n=90 · TNBC n=80

Evidence: Methods, pp. 2-3

From biomarkers to signature

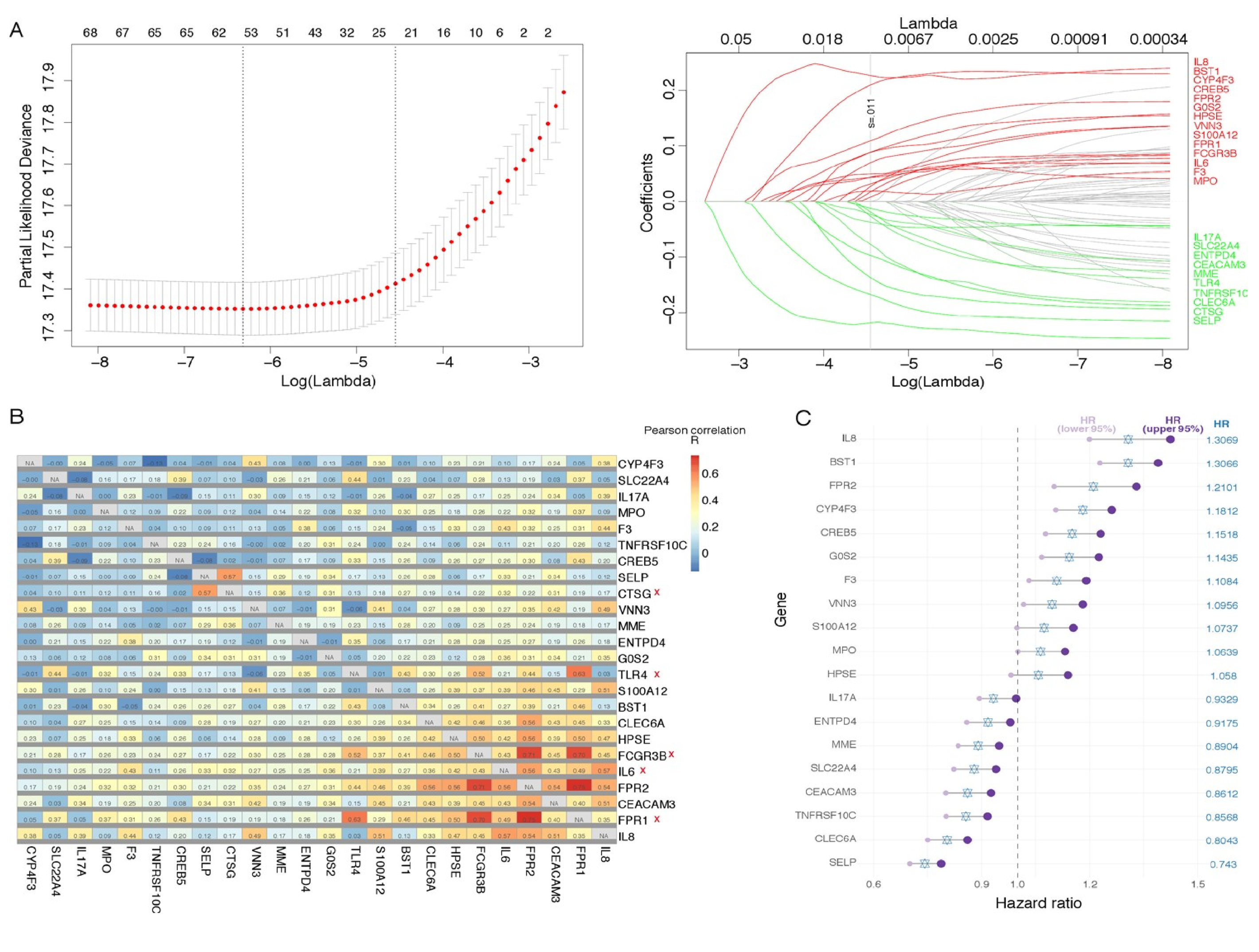

Figure 1 · Construction of the pan-cancer NETs signature
Source paper, p. 4 · original colors locked

Higher NETs score identified patients with poorer survival

Worse DSS, OS and PFI across most cancer types

Evidence: Results, p. 4; Figure 2

Pan-cancer prognostic performance

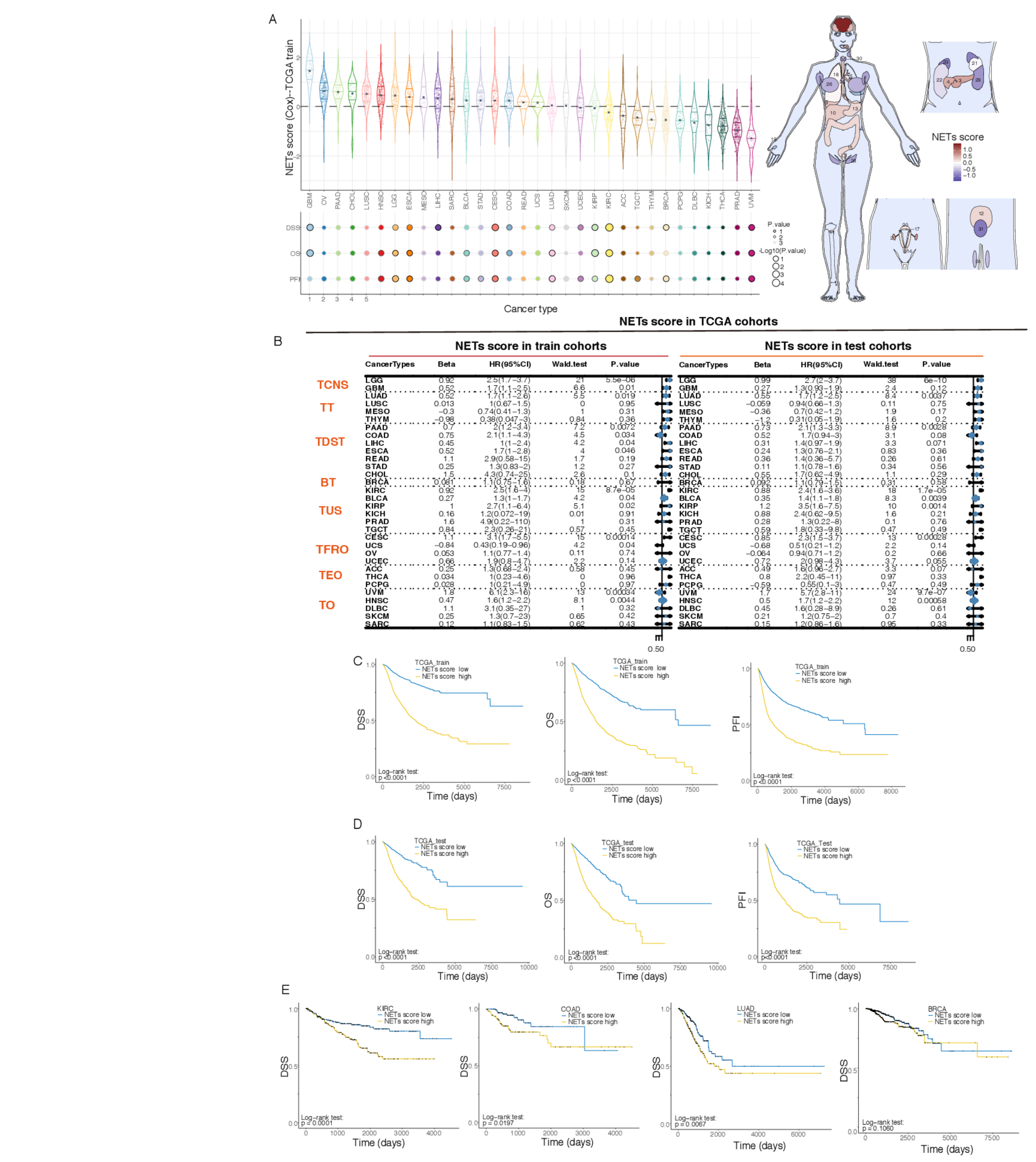

Figure 2 · NETs score distribution and survival performance
Source paper, p. 5 · original colors locked

Independent and external validation

- TCGA test cohort: OS HR=1.22, p=0.0070 after adjustment for cancer type.
- External cohorts supported prognostic value in glioma and LUAD.
- Results were not consistent in breast cancer (7/8 cohorts non-significant) or LUSC.
- Tumor-type context remains essential for interpretation.

Evidence: Results, pp. 4-6

Nomogram AUC improved: 0.700→0.726 (train) · 0.661→0.718 (test)

Malignant biology

R=0.7444 Epithelial-mesenchymal transition

R=0.5369 Angiogenesis

R=0.3835 Cell cycle

All correlations p<0.0001 · Results, p. 6

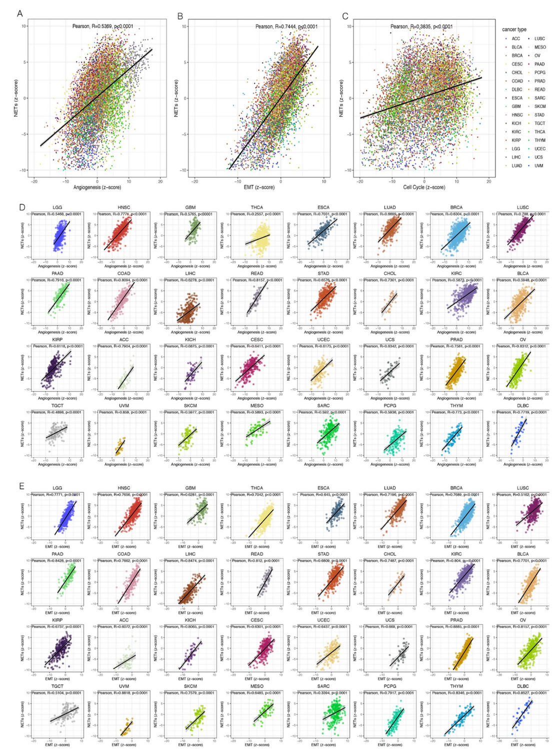

Figure 5 · NETs activity aligns with aggressive tumor programs
Source paper, p. 8 · original colors locked

Clinical IHC validation

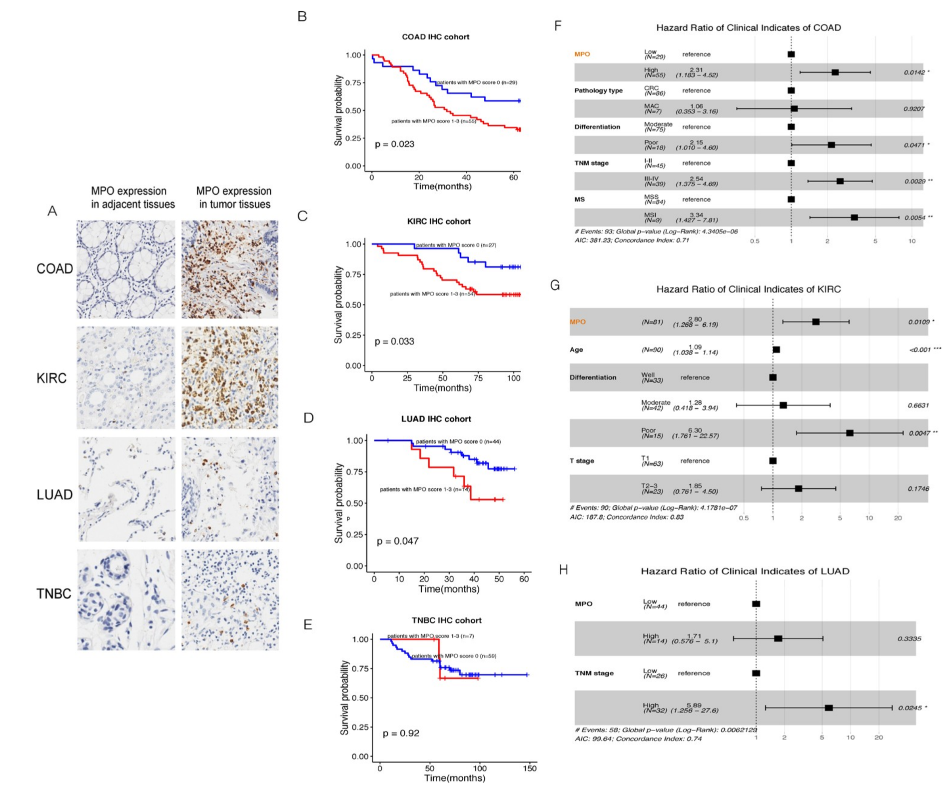

Figure 6 · MPO stratified outcomes in COAD, KIRC and LUAD, not TNBC
Source paper, p. 9 · original colors locked

Conclusion

A 19-gene transcriptomic signature captured clinically relevant NETs-formation potential, prognosis and malignant biology across cancers.

Limitations: few NETs markers, limited cancer-specific IHC cohorts and insufficient H3Cit positivity; broader multi-marker validation is required.

Evidence: Discussion and Conclusions, p. 10

Zhang et al. J Immunother Cancer. 2022;10:e004210 · CC BY-NC · traceable paraphrase

Approvals current · scientific figure colors locked